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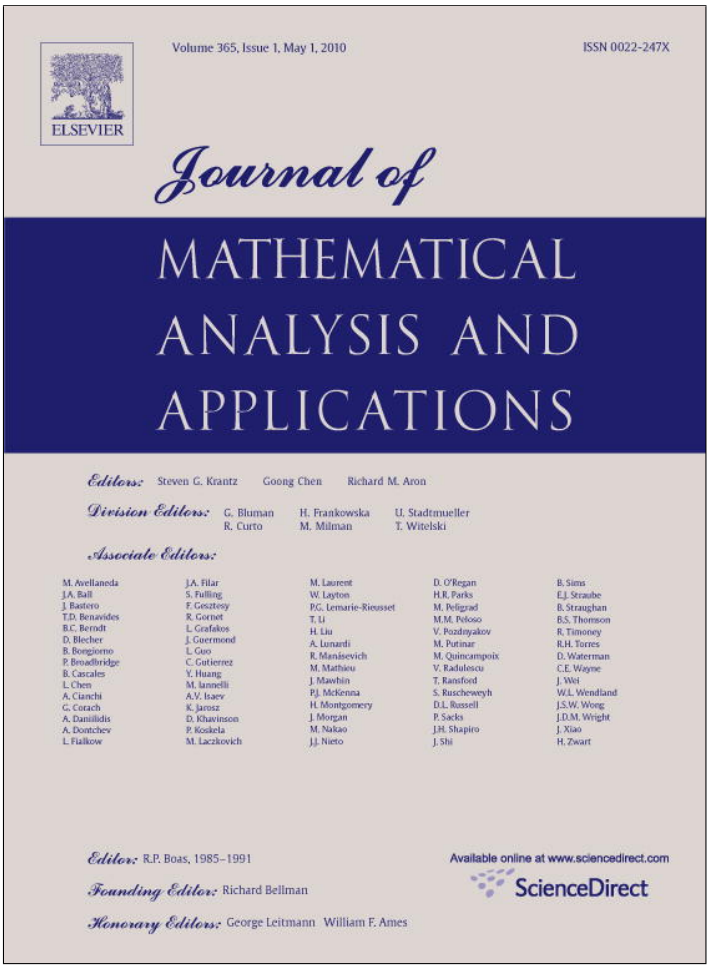

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De Finetti-type theorems for random selection processes. Necessary and sufficient conditions

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ABSTRACT

This paper is concerned with random selection processes for which, in the previous literature, several sufficient conditions for a De Finetti-type theorem (granting the conditional independence of a family of random variables) have been proposed. In this paper, we prove that these conditions are also necessary. In this sense, we identify the extremes of the convex set of marginal distributions of a random selection process, and then we give a thorough description and classification of such processes.

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1. Introduction

In the probabilistic literature, the De Finetti theorem, which ensures the conditional independence of a family of random variables, is always related to exchangeability; see, for instance [1–3,6]. There are, however, more general settings allowing to obtain the conditional independence of a family of random variables, as in De Finetti's theorem, without this exchangeability assumption and, particularly, for non-identically distributed random variables.

In our quest of De Finetti-type results on the conditional independence of non-identically distributed random variables, we considered in [10,11] the notion of a *random selection process*, which is defined as follows: Given a nonempty set E and any weight function $m : E \to (0, \infty)$, a random selection process $\boldsymbol{N}$ on (E, m) is a family of random variables

$$N(x) \in \{0, 1\} \quad \text{for } x \in E$$

on a probability space $(\Omega, \mathcal{F}, \mathrm{P})$ verifying, for every finite subset $B \subseteq E$, the following conditions:

(C1) Conditional on $N(B) := \sum_{x \in B} N(x)$, $\{N(x)\}_{x \in B}$ and $\{N(x)\}_{x \in B^c}$ are independent.
(C2) $\mathrm{P}\{N(x) = 1 \mid N(B) = 1\} = m(x)/m(B)$ for all $x \in B$, where $m(B) := \sum_{x \in B} m(x)$.

We can interpret that $N(x) = 1$ means that the element $x \in E$ is "selected", while $x \in E$ is "rejected" when $N(x) = 0$. The Markovian condition (C1) means that, once the number of points to be selected in B is known, the points selected inside B do not depend on the selections made outside B. The condition (C2) establishes a preference relation among the points of any set B; in the sense that, if only a point in B is to be selected, each one has probability proportional to its weight. It can be shown that the exchangeability of $\{N(x)\}_{x \in E}$ is equivalent to (C1) and (C2) with $m(x) \equiv 1$. This is shown in Proposition 1.1 in [11], which we restate for ease of reference. (Let us mention that, in Section 4, we propose an example of a random selection process with non-constant weight function m.)

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Proposition 1.1. *A family of exchangeable indicator random variables $\{N(x)\}_{x\in E}$ is a random selection process with constant weight function.*

Conversely, if $\boldsymbol{N}$ is a random selection process on (E,m) with constant weight function, then the random variables $\{N(x)\}_{x\in E}$ are exchangeable.

Thus, the usual exchangeability condition splits in the Markovian character (C1) plus the stipulation (C2) of uniformity among the points of any set. In [11], it is proved that De Finetti-like theorems hold provided that the weight function m satisfies some assumptions to be specified later. Here we want to show that these conditions are not only sufficient, but also necessary to grant the conditional independence of the random variables $\{N(x)\}_{x\in E}$.

First of all, let us introduce some notation. Let $\mathcal{B}$ be the ring of all the finite subsets of E. For $B\in\mathcal{B}$ and $0\leqslant r\leqslant \#B$ (here, $\#B$ denotes the cardinal of the set B), define

$$M_r(B) := \sum_{x_1,\dots,x_r\in B}\prod_{i=1}^{r} m(x_i),$$

where the sum extends over all the subsets of B with r elements. For ease of notation, we let $M_r(B):=0$ if $r<0$ or $r>\#B$. The values $M_r(B)$ may be alternatively determined as the successive coefficients of the polynomial $\prod_{x\in B}[1+m(x)z]$.

It is easily seen that the conditional distributions of a random selection process are necessarily of the hypergeometric type [10, Proposition 2.3]; that is, given finite sets $B_1\subseteq B$ and $1\leqslant r\leqslant \#B$, the conditional distribution of $N(B_1)$ given $N(B)=r$ is the "generalized hypergeometric" distribution:

$$\mathrm{P}\big\{N(B_1)=s \bigm| N(B)=r\big\} = \frac{M_s(B_1)M_{r-s}(B-B_1)}{M_r(B)}. \tag{1.1}$$

As a direct consequence, if $B_1,\dots,B_k$ is a partition of $B\in\mathcal{B}$, then

$$\mathrm{P}\big\{N(B_1)=s_1,\dots,N(B_k)=s_k \bigm| N(B)=r\big\} = \frac{M_{s_1}(B_1)\cdots M_{s_k}(B_k)}{M_r(B)}$$

if $r=\sum s_i$. This shows that the distribution of $\boldsymbol{N}$ inside each finite set B is determined by the marginal distribution of $N(B)$. Now, the main issue is how to specify a family of *compatible marginal distributions* for all the sets $B\in\mathcal{B}$, in the sense that the "total probability rule":

$$\mathrm{P}\big\{N(B)=s\big\} = \sum_{r=0}^{\#B'}\mathrm{P}\big\{N\big(B\cup B'\big)=s+r\big\}\frac{M_s(B)M_r(B')}{M_{s+r}(B\cup B')} \tag{1.2}$$

is verified for every disjoint sets $B,B'\in\mathcal{B}$; see [10, Lemma 2.4]. Clearly, if E is a finite set, then we can arbitrarily choose the distribution of $N(E)$ on $\{0,1,\dots,\#E\}$ and then deduce the marginal distributions of the random selection process by means of (1.1), by replacing B with E. Therefore, in the forthcoming, we will suppose that *the set E is not finite*.

As shown in [10, Theorem 2.5], an obvious universal solution for (1.2), which is valid for any set E and any weight function m, is to take as marginal distributions any mixture of the "generalized binomial" distributions: for each finite set $B\in\mathcal{B}$ let

$$\mathrm{P}\big\{N(B)=s\big\} = \int_0^1 \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+\xi m(x)]}F(d\xi) \quad \text{for } 0\leqslant s\leqslant \#B, \tag{1.3}$$

where F is any fixed probability distribution on $[0,1]$. In this case, the random variables $\{N(x)\}_{x\in E}$ are conditionally independent given ξ, with

$$\mathrm{P}\big\{N(x)=1 \bigm| \xi\big\} = \frac{\xi m(x)}{1-\xi+\xi m(x)} \quad \text{and} \quad \mathrm{P}\big\{N(x)=0 \bigm| \xi\big\} = \frac{1-\xi}{1-\xi+\xi m(x)}. \tag{1.4}$$

The main results proved in [11] are as follows. Suppose that there exists a denumerable set $A:=\{x_q\}_{q\geqslant 1}\subseteq E$ such that either:

(A1) The sequence $\{m(x_q)\}$ converges to some $m\in(0,\infty)$, or
(A2) The sequence $\{m(x_q)\}$ converges to zero and, in addition, $m(A):=\sum m(x_q)=\infty$, or
(A3) The sequence $\{m(x_q)\}$ diverges to ∞ and, in addition, $\hat{m}(A):=\sum 1/m(x_q)=\infty$.

Then there exists a distribution function F on $[0,1]$ such that (1.3) holds for every $B\in\mathcal{B}$. Consequently, the random variables $\{N(x)\}_{x\in E}$ are conditionally independent given ξ, and (1.4) holds. These De Finetti-type theorems are obtained from a strong law of large numbers, ensuring the existence of the almost sure limit, as $n\to\infty$, of

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$$\frac{N(A_n)}{n}, \qquad \frac{N(A_n)}{m(A_n)} \quad \text{or} \quad \frac{n-N(A_n)}{\hat{m}(A_n)} \tag{1.5}$$

for the cases (A1), (A2), and (A3), respectively, where $A_n := \{x_1, \dots, x_n\}$. For details, see [11]. (We note that the sufficient conditions are stated in [11] in a different, though equivalent, form.)

After having established the preceding results, we were advised by an anonymous referee that random selection processes (for denumerable E) are closely related to generalized Stirling triangles of the first kind, as defined by Kerov in [8, Section 1.4]. Moreover, the above mentioned case (A3) appears as a conjecture made by Kerov in his previous paper [7]. See [10, Section 4] for a summary of the relation between random selection processes and generalized Stirling triangles.

As already mentioned, in this paper we will prove that the conditions (A1), (A2), and (A3) are not only *sufficient* to obtain a De Finetti-type theorem, but they are also *necessary*. More precisely, we will show that if none of these assumptions hold, then there exist compatible marginal distributions (satisfying (1.2)) different from those in (1.3), and so the conditional independence of the random variables $\{N(x)\}_{x\in E}$ is not granted.

Equivalently, let $\mathcal{P}$ be the set of all functions $p(s, B)$, for $B \in \mathcal{B}$ and $s = 0, 1, \dots, \#B$, such that $p(\cdot, B)$ is a probability distribution on $\{0, 1, \dots, \#B\}$ for each $B \in \mathcal{B}$ and, in addition, (1.2) is verified (replacing $\mathrm{P}\{N(B) = s\}$ with $p(s, B)$). Clearly, $\mathcal{P}$ is a compact convex set within a locally convex vector space endowed with the topology of the pointwise convergence. Therefore, $\mathcal{P}$ is the closed convex hull of its extreme points $\hat{\mathcal{P}}$.

Under the hypotheses (A1)–(A3) above, it is proved in [11] that $\mathcal{P}$ is the set of the functions given by the right member of (1.3), and $\hat{\mathcal{P}}$ is the family

$$\hat{p}_\xi(s, B) := \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+\xi m(x)]} \quad \text{for } B \in \mathcal{B} \text{ and } 0 \leqslant s \leqslant \#B, \tag{1.6}$$

as ξ varies in $[0, 1]$. The main contribution of this paper is to identify the set $\hat{\mathcal{P}}$ of extreme points when (A1)–(A3) are not verified. We will show that they are, then, different from (1.6), and so a De Finetti-type theorem does not hold. Interestingly, in this case, $\hat{\mathcal{P}}$ is denumerable, as opposed to the previous situation, despite of the fact that (1.6) always belongs to $\mathcal{P}$ for any $\xi \in [0, 1]$.

To bring Kerov's analysis to our context, we will say that a function $\varphi(s, B) \geqslant 0$ is harmonic (in E with respect to m) if it verifies

$$\varphi(s, B) = \sum_{r=0}^{\#B'} \varphi\big(s+r, B\cup B'\big) M_r\big(B'\big) \quad \text{and} \quad \varphi(0, \emptyset) = 1.$$

Clearly $\varphi(s, B) := p(s, B)/M_s(B)$ establishes an isomorphism between $\mathcal{P}$ and the compact convex set of positive harmonic functions $\mathcal{V}$. Thus, the problem of finding the extremes of $\mathcal{V}$ is equivalent to the determination of $\hat{\mathcal{P}}$.

The rest of the paper is organized as follows. In Section 2 we show some preliminary useful results on random selection processes. In Section 3 we introduce the outstanding alternatives to (A1)–(A3), and we determine the corresponding extreme points $\hat{\mathcal{P}}$ of the family of compatible marginal distributions (this is done in Sections 3.1 and 3.2). In Section 3.3 we make some comments on the so-called rejection function of the random selection process. An example of a random selection process is proposed in Section 4. Finally, the conclusions are given in Section 5.

2. Preliminary results on random selection processes

Associated to a random selection process $\boldsymbol{N}$ on (E, m), it will be useful to consider $\hat{\boldsymbol{N}}$ given by

$$\hat{N}(x) := 1 - N(x) \quad \text{and} \quad \hat{N}(B) := \sum_{x\in B} \hat{N}(x) = \#B - N(B) \tag{2.1}$$

for all $x \in E$ and all $B \in \mathcal{B}$. Thus, $\hat{N}$ gives the number of rejected points in each set B and, obviously, it verifies (C1) and (C2) since

$$\mathrm{P}\big\{\hat{N}(x) = 1 \,\big|\, \hat{N}(B) = 1\big\} = \mathrm{P}\big\{N(x) = 0 \,\big|\, N(B) = \#B - 1\big\} = \frac{M_{\#B-1}(B-x)}{M_{\#B-1}(B)} = \frac{1/m(x)}{\sum_{y\in B} 1/m(y)}.$$

Therefore, $\hat{\boldsymbol{N}}$ is a random selection process on $(E, \hat{m})$, with

$$\hat{m}(x) := 1/m(x) \quad \text{and} \quad \hat{m}(B) := \sum_{x\in B} \hat{m}(x).$$

By $\hat{M}_r(B)$ we will denote the coefficients of the polynomial $\prod_{x\in B}[1 + z\hat{m}(x)]$ and, as easily seen, it is, for all r,

$$\hat{M}_{\#B-r}(B) = \frac{M_r(B)}{\prod_{x\in B} m(x)}. \tag{2.2}$$

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The foregoing simple observation shows that, among the results cited in Section 1, the case (A3) is merely a transcription of the case (A2) for the random selection process $\hat{\boldsymbol{N}}$.

Let us begin with a slight modification of the result in [11, Lemma 3.1].

Lemma 2.1. *Let $\boldsymbol{N}$ be a random selection process on (E, m), let $A := \{x_q\}_{q\geqslant 1}$ be any denumerable set in E and $A_n := \{x_q\}_{1\leqslant q\leqslant n}$ for $n \geqslant 1$. Consider the reverse filtration*

$$\mathcal{G}_n^A := \sigma\big(N(A_n), N(A_{n+1}), \ldots\big) = \sigma\big(N(A_n), N(x_{n+1}), N(x_{n+2}), \ldots\big).$$

Then, for any finite set $B \subset A$ and any $0 \leqslant s \leqslant \#B$, the reverse martingale

$$Z_n^A(s, B) := \mathrm{P}\big\{N(B) = s \mid \mathcal{G}_n^A\big\}$$

converges almost surely to

$$Z_\infty^A(s, B) := \mathrm{P}\big\{N(B) = s \mid \mathcal{G}_\infty^A\big\},$$

where $\mathcal{G}_\infty^A := \bigcap_n \mathcal{G}_n^A$. Moreover, if n is large enough so that $B \subseteq A_n$, then

$$Z_n^A(s, B) = \left[\frac{M_s(B)M_{r-s}(A_n - B)}{M_r(A_n)}\right]_{r=N(A_n)}. \tag{2.3}$$

This lemma mainly ensures that, for all finite $B \subset A$ and $0 \leqslant s \leqslant \#B$, the limits

$$\lim_n \left[\frac{M_{r-s}(A_n - B)}{M_r(A_n)}\right]_{r=N(A_n)}$$

exist with probability one[1] and they almost surely yield a probability distribution when multiplied by $M_s(B)$. This probability distribution can be called the *tail conditioned distribution of $N(B)$ relative to A*.

For a fixed denumerable set $A \subseteq E$, let $\bar{\Omega}(A)$ be the set where $Z_\infty^A(1, x)$ and $Z_\infty^A(0, x)$ exist and $Z_\infty^A(0, x) + Z_\infty^A(1, x) = 1$ for all $x \in A$. According to Lemma 2.1, it is $\mathrm{P}(\bar{\Omega}(A)) = 1$.

Proposition 2.2. *Let $\boldsymbol{N}$ be a random selection process on (E, m) and let $A := \{x_q\}_{q\geqslant 1}$ be any denumerable set in E. Then, up to sets of probability 0,*

$$\big\{N(A) = 0\big\} = \big\{Z_\infty^A(1, x) = 0,\ \forall x \in A\big\} = \big\{Z_\infty^A(1, x_1) = 0\big\},$$
$$\big\{\hat{N}(A) = 0\big\} = \big\{Z_\infty^A(0, x) = 0,\ \forall x \in A\big\} = \big\{Z_\infty^A(0, x_1) = 0\big\}.$$

Proof. Inside $\bar{\Omega}(A)$, if $N(A) = 0$, then $Z_\infty^A(1, x) = 0$ for all $x \in A$ since $[M_{r-1}(A_n - x)]_{r=N(A_n)} = 0$. Similarly, if $\hat{N}(A) = 0$ or $N(A_n) = n$ for all $n \geqslant 1$, it is $M_{N(A_n)}(A_n - x) = 0$ and $Z_\infty^A(0, x) = 0$ for all $x \in A$. Moreover, for any $x \in A$,

$$Z_\infty^A(1, x) = \frac{m(x)}{m(x_1)} Z_\infty^A(1, x_1) \lim_n \left[\frac{M_{r-1}(A_n - x)}{M_{r-1}(A_n - x_1)}\right]_{r=N(A_n)}$$

so that the last limit exists and is bounded between $m(x_1)/m(x) \wedge 1$ and $m(x_1)/m(x) \vee 1$ (because each term in the sum $M_{r-1}(A_n - x)$ is obtained from a term of $M_{r-1}(A_n - x_1)$ either directly or by multiplying by $m(x_1)/m(x)$). Assume that, for some $\omega \in \bar{\Omega}(A)$, it is $Z_\infty^A(1, x_1) = 0$ while $N(A) > 0$; then $Z_\infty^A(1, x) = 0$ for all $x \in A$, and this means

$$\mathrm{P}\big\{N(x) = 1 \mid \mathcal{G}_\infty^A\big\} = 0, \quad \forall x \in A, \quad \text{or} \quad \mathrm{P}\big\{N(A) > 0 \mid \mathcal{G}_\infty^A\big\} = 0$$

(out of the negligible set where the last probability does not match the sum of the preceding ones). But $\{N(A) = 0\} \in \mathcal{G}_\infty^A$, and hence $\mathbf{I}_{\{N(A)>0\}}$ is a version of the last conditional probability, taking the value 1 at ω. Thus $N_1(A) := \bar{\Omega}(A) \cap \{Z_\infty^A(1, x_1) = 0,\ N(A) > 0\}$ is a set of probability 0. Also

$$Z_\infty^A(0, x) = Z_\infty^A(0, x_1) \lim_n \left[\frac{M_r(A_n - x)}{M_r(A_n - x_1)}\right]_{r=N(A_n)}$$

and $N_0(A) := \bar{\Omega}(A) \cap \{Z_\infty^A(0, x_1) = 0,\ \hat{N}(A) > 0\}$. Similarly, we obtain $\mathrm{P}(N_0(A)) = 0$. $\square$

[1] This conclusion is related with [4, Lemma 5] where a similar result follows, in more general circumstances, from the theory of the Martin boundary associated to some Markov chain. It can also be linked with the theorem in [8, p. 60] whose proof is an adaptation of Birkhoff's pointwise ergodic theorem. See also [10, Section 4].

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Exclude from $\bar{\Omega}(A)$ the anomalous negligible sets $N_1(A)$ and $N_0(A)$; from now on $\bar{\Omega}(A)$ will denote the resulting set of probability 1. Moreover, $\Omega_0(A) := \bar{\Omega}(A) \cap \{N(A) = 0\}$ will represent the set of trajectories in $\bar{\Omega}(A)$ that select no element of A; $\Omega_1(A) := \bar{\Omega}(A) \cap \{\hat{N}(A) = 0\}$ will denote the trajectories that select all the elements of A; while we let $\Omega^\star(A) := \bar{\Omega}(A) - (\Omega_0(A) \cup \Omega_1(A))$.

In $\Omega^\star(A)$, for any $x \in A$, it is $Z^A_\infty(1,x) \in (0,1)$ and

$$Z^A_\infty(1,x) = m(x) Z^A_\infty(0,x) \lim_n \left[\frac{M_{r-1}(A_n - x)}{M_r(A_n - x)} \right]_{r=N(A_n)}.$$

Therefore, the last limit exists and has a value $\theta^A(x) \in (0,\infty)$ that allows to define $\xi^A(x) := \theta^A(x)/(1+\theta^A(x)) \in (0,1)$ in order to get

$$Z^A_\infty(1,x) = \frac{m(x)\xi^A(x)}{1 - \xi^A(x) + m(x)\xi^A(x)} \quad \text{and} \quad Z_\infty(0,x) = \frac{1 - \xi^A(x)}{1 - \xi^A(x) + m(x)\xi^A(x)}. \tag{2.4}$$

Thus $\Omega_0(A) = \{\xi^A(x) = 0\}$ for all $x \in A$, and $\Omega_1(A) = \{\xi^A(x) = 1\}$ for all $x \in A$.

When E is denumerable, Proposition 2.2 may be applied to E itself, in order to consider $\Omega_0(E) = \bar{\Omega}(E) \cap \{N(E) = 0\}$, $\Omega_1(E) = \bar{\Omega}(E) \cap \{\hat{N}(E) = 0\}$ and $\Omega^\star(E) = \bar{\Omega}(E) - (\Omega_0(E) \cup \Omega_1(E))$. Then (2.4) holds for all $x \in E$ (with $\xi^A(x)$ replaced with $\xi^E(x)$).

Lemma 2.3. *Let $A := \{x_q\}_{q \geqslant 1}$ be any denumerable subset of E. If $r = r(n)$ are integers such that $0 \leqslant r \leqslant n$ for all n and $0 < r < n$ for all large n, then, for any fixed $x \in A$,*

$$\frac{r}{m(A_n - x) - \min_B m(B)} \leqslant \frac{M_{r-1}(A_n - x)}{M_r(A_n - x)} \leqslant \frac{r}{m(A_n - x) - \max_B m(B)} \tag{2.5}$$

where $A_n := \{x_1, \ldots, x_n\}$, and B is any subset of $A_n - x$ with $\#B = r - 1$.

Proof. Let us observe that

$$\sum_B \prod_{x \in B} m(x) m(A_n - x - B) = r M_r(A_n - x),$$

where the sum extends over all subsets B of $A_n - x$ with $\#B = r - 1$; therefore

$$M_{r-1}(A_n - x) \min_B m(A_n - x - B) \leqslant r M_r(A_n - x) \leqslant M_{r-1}(A_n - x) \max_B m(A_n - x - B)$$

and (2.5) follows. □

Proposition 2.4. *Let $\boldsymbol{N}$ be a random selection process on (E,m), and let A any sequence in E such that $m(A) < \infty$. Then, with probability 1, $\hat{N}(A) = 0$ or $N(A) < \infty$.*

Proof. In $\Omega^\star(A)$ we may apply Lemma 2.3 with $r = N(A_n)$ and, for any $x \in A$, we get

$$\frac{N(A_n)}{m(A_n)} \leqslant \left[\frac{M_{r-1}(A_n - x)}{M_r(A_n - x)} \right]_{r=N(A_n)}.$$

But the right member has a limit $\theta^A(x) \in (0,\infty)$ and $m(A_n) \uparrow m(A) < \infty$; thus $N(A) < \infty$. □

3. The alternative to De Finetti's theorem

In this section, we are going to determine the set of compatible marginal distributions of a random selection process $\boldsymbol{N}$ on (E,m) when the assumptions (A1)–(A3) in Section 1 do not hold. It is worthwhile to note that the hypothesis (A1) is verified whenever E is an uncountable set (see [11, Section 5]); thus, in the sequel, we will assume that *E is denumerable*.

Given an arbitrary constant $c > 0$, let us consider

$$E^0_c := \{x \in E\colon m(x) \leqslant c\} \quad \text{and} \quad E^\infty_c := \{x \in E\colon m(x) > c\}.$$

If the assumptions (A1)–(A3) do not hold, then one of the following outstanding alternatives necessarily holds:

(H1) $\{m(x)\}_{x \in E}$ has only the limit point 0 and $m(E) < \infty$.
(H2) $\{m(x)\}_{x \in E}$ has only the limit point ∞ and $\hat{m}(E) < \infty$.
(H3) $\{m(x)\}_{x \in E}$ has only two limit points, 0 and ∞. Thus, for every $c > 0$, E^0_c and E^∞_c are denumerably infinite and, in addition, $m(E^0_c) < \infty$ and $\hat{m}(E^\infty_c) < \infty$.

The case of the conditions (H1) and (H2) is analyzed in Section 3.1 below, while (H3) is studied in Section 3.2.

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3.1. Random selection processes under assumptions (H1) or (H2)

As a first consequence of assumption (H1), the infinite product $\prod_{x\in E}[1+zm(x)]$ is convergent and, therefore, $M_r(B)$ is finite for any $r \geqslant 0$ and any (not necessarily finite) set $B \subseteq E$.

On the other hand, assumption (H1) allows to apply Proposition 2.4 to E itself, in order to conclude that $\hat{N}(E)=0$ or $N(E)<\infty$. Thus we can consider the distribution of $N(E)$, extended to $\mathbb{N}\cup\{\infty\}$, in the following sense:

$$\gamma(u) := \mathrm{P}\big\{N(E)=u\big\} \quad \text{for } u \geqslant 0, \quad \text{and} \quad \gamma(\infty) := \mathrm{P}\big\{\hat{N}(E)=0\big\}.$$

Now, a simple limit argument, starting from (1.1), shows that, for any $B \subseteq E$,

$$\mathrm{P}\big\{N(B)=s \bigm| N(E)=u\big\} = \frac{M_s(B)M_{u-s}(E-B)}{M_u(E)} =: \mathcal{H}(u;s,B) \quad \text{for } 0 \leqslant s \leqslant \#B \wedge u.$$

Here, $\mathcal{H}$ denotes the "generalized hypergeometric" distribution associated to the weights $m(x)$. Therefore, the marginal distributions of $N(B)$, for $B \in \mathcal{B}$, are given by

$$\mathrm{P}\big\{N(B)=s\big\} = \sum_{u=s}^{\infty} \gamma(u)\mathcal{H}(u;s,B) + \gamma(\infty)\mathbf{I}_{\{s=\#B\}} \tag{3.1}$$

and they belong to the closed convex hull generated by the $\{\mathcal{H}(u;s,B)\}_{u\in\mathbb{N}\cup\{\infty\}}$, where $\mathcal{H}(\infty;s,B)$ is the causal distribution on $s=\#B$. In fact, it is easy to check that, for any fixed u, $\mathcal{H}(u;\cdot,\cdot)$ satisfies (1.2) and, thus, belongs to $\mathcal{P}$. Moreover, it is an extreme of $\mathcal{P}$, since from two elements of $\mathcal{P}$ attributing probability less than one to $\{N(E)=u\}$, one cannot get a mixture of them which gives probability one to this set.

In summary, we have established the following statement.

Theorem 3.1. *Let $\boldsymbol{N}$ be a random selection process on (E,m) and assume that* (H1) *is verified. Then, the marginal distributions for $\{N(B)\}_{B\in\mathcal{B}}$ are given by* (3.1)*, where $\gamma(u)$ may be any probability distribution on $\mathbb{N}\cup\{\infty\}$. They are, thus, mixtures of the extreme "generalized hypergeometric" distributions*: $\{\mathcal{H}(u;s,B)\}_{u\in\mathbb{N}\cup\{\infty\}}$.

The case of assumption (H2) is a corollary of Theorem 3.1; it suffices to consider the random selection process $\hat{\boldsymbol{N}}$ on $(E,\hat{m})$, that satisfies (H1), and to apply Theorem 3.1. As it will be $N(E)=0$ or $\hat{N}(E)<\infty$, let us consider

$$\hat{\gamma}(v) := \mathrm{P}\big\{\hat{N}(E)=v\big\} \quad \text{for } u \geqslant 0, \qquad \hat{\gamma}(\infty) := \mathrm{P}\big\{N(E)=0\big\},$$

and

$$\hat{\mathcal{H}}(v;s,B) := \frac{\hat{M}_{\#B-s}(B)\hat{M}_{v+s-\#B}(E-B)}{\hat{M}_v(E)},$$

that are now the extreme elements of $\mathcal{P}$.

Theorem 3.2. *Let $\boldsymbol{N}$ be a random selection process on (E,m) and assume that* (H2) *is verified. Then, the marginal distributions for $\{N(B)\}_{B\in\mathcal{B}}$ are given by*

$$\mathrm{P}\big\{N(B)=s\big\} = \sum_{v=\#B-s}^{\infty} \hat{\gamma}(v)\hat{\mathcal{H}}(v;s,B) + \hat{\gamma}(\infty)\mathbf{I}_{\{s=0\}} \tag{3.2}$$

where $\hat{\gamma}(v)$ may be any probability distribution on $\mathbb{N}\cup\{\infty\}$. They are, thus, mixtures of the extreme "generalized hypergeometric" distributions: $\{\hat{\mathcal{H}}(v;s,B)\}_{v\in\mathbb{N}\cup\{\infty\}}$.

This last result contains Kerov's assertion [8, Theorem 5, p. 71] for the case $\hat{m}(E)<\infty$.

For a random selection process in (E,m) such that one of the assumptions (A1), (A2), or (A3) is satisfied, Theorems 3.3 and 3.5 in [11] ensure that $\mathcal{P}$ is the family of marginal distributions given by (1.3) for some distribution F in $[0,1]$, whose uncountable extremes are the "generalized binomial" distributions given in (1.6). Under hypothesis (H1) or (H2), when $\mathcal{P}$ is the closed convex hull generated by the countable extremes $\{\mathcal{H}(u;s,B)\}_{u\in\mathbb{N}\cup\{\infty\}}$ or $\{\hat{\mathcal{H}}(v;s,B)\}_{v\in\mathbb{N}\cup\{\infty\}}$, it is interesting to examine how the "generalized binomial" distributions arise.

In the case that (H1) holds, for each $\theta \in [0,\infty)$ we let

$$\gamma_\theta(u) := \frac{M_u(E)\theta^u}{\prod_{x\in E}[1+m(x)\theta]} \quad \text{and} \quad \gamma_\theta(\infty) := 0. \tag{3.3}$$

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Then, (3.1) gives, for any $B \in \mathcal{B}$,

$$\begin{aligned}\mathrm{P}\{N(B)=s\} &= \sum_{u=s}^{\infty} \frac{\theta^u}{\prod_{x\in E}[1+m(x)\theta]} M_s(B) M_{u-s}(E-B) \\ &= \frac{M_s(B)\theta^s}{\prod_{x\in E}[1+m(x)\theta]} \sum_{t=0}^{\infty} \theta^t M_t(E-B) \\ &= \frac{M_s(B)\theta^s}{\prod_{x\in B}[1+m(x)\theta]}\end{aligned}$$

that coincides with (1.6) if θ is replaced with $\xi/(1-\xi)$, for some $\xi \in [0,1)$. The case $\xi = 1$ is obviously obtained when $\gamma(\infty) = 1$. When ξ is a random variable on $[0,1]$, the mixture (1.3) corresponds to $\gamma(u) = \mathrm{E}_\theta[\gamma_\theta(u)]$. Of course, the random variables $\{N(x)\}_{x\in E}$ are then conditionally independent, given ξ (or θ).

Similarly, for the case in which (H2) holds, if $\theta \in [0,\infty)$ and $\hat{\gamma}(v)$ is given by:

$$\hat{\gamma}_\theta(v) := \frac{\hat{M}_v(E)\theta^v}{\prod_{x\in E}[1+\hat{m}(x)\theta]}, \qquad \hat{\gamma}_\theta(\infty) := 0, \tag{3.4}$$

for any $B \in \mathcal{B}$, we get from (3.2)

$$\mathrm{P}\{N(B)=s\} = \frac{\hat{M}_{\#B-s}(B)\theta^{\#B-s}}{\prod_{x\in B}[1+\hat{m}(x)\theta]} = \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+m(x)\xi]}$$

after using (2.2) and once θ is replaced with $(1-\xi)/\xi$ for $\xi \in (0,1]$. For $\hat{\gamma}(\infty) = 1$ we get the same result with $\xi = 0$. As before, a random choice of $\xi \in [0,1]$ corresponds to $\hat{\gamma}(v) = \mathrm{E}_\theta[\hat{\gamma}_\theta(v)]$ and the random variables $\{N(x)\}_{x\in E}$ are conditionally independent given ξ (or θ).

Let us point out some further differences between the case considered here and the results obtained in [11]. When some of the assumptions (A1), (A2), or (A3) are fulfilled, it can be shown that $\xi^A(x)$ in (2.4) has a random value ξ, independent of x and A. Moreover $\mathcal{G}_\infty^A$ is the completion of the σ-field $\sigma(\xi)$, for any sequence A, and

$$Z_\infty^A(s,B) = \mathrm{P}\{N(B)=s \mid \xi\} = \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+\xi m(x)]}$$

depends only on the weights in B. On the contrary, under assumption (H1), for any sequence $A \subseteq E$ it is $\mathcal{G}_\infty^A = \sigma(N(A))$ (recall that $\{N(A)=\infty\} \overset{\text{a.s.}}{=} \{\hat{N}(A)=0\}$) and, according to (2.3),

$$Z_\infty^A(s,B) = M_s(B)\left[\frac{M_{r-s}(A-B)}{M_r(A)}\right]_{r=N(A)} \quad \text{for } B \subseteq A. \tag{3.5}$$

These conditional distributions depend on A and on all the weights in A. When $A = E$, the same is true for their expectations, given by (3.1), except if $\gamma = \gamma_\theta$ with a fixed or random value of θ. Note also that the σ-field $\mathcal{G}_\infty^E$ is trivial under each extreme $\mathcal{H}(u;s,B)$, since $N(E) \overset{\text{a.s}}{=} u$.

Assuming (H2), an analogous result holds: $\mathcal{G}_\infty^A = \sigma(\hat{N}(A))$ and

$$Z_\infty^A(\#B-s,B) = \hat{M}_s(B)\left[\frac{\hat{M}_{r-s}(A-B)}{\hat{M}_r(A)}\right]_{r=\hat{N}(A)} \quad \text{for } B \subseteq A. \tag{3.6}$$

3.2. Random selection processes under assumption (H3)

As a first consequence of assumption (H3), the infinite products

$$\prod_{x\in E_c^0}\left[1+zm(x)\right] \quad \text{and} \quad \prod_{x\in E_c^\infty}\left[1+z\hat{m}(x)\right]$$

are convergent for any value of $c > 0$. Therefore, we can define $M_r(B)$ for any $B \subseteq E_c^0$ and $\hat{M}_r(B)$ for any $B \subseteq E_c^\infty$ as the respective coefficients of z^r, which are finite, in the corresponding infinite products. Moreover, it will be convenient to think of the elements of E as partially ordered according to their weights, and to think of the index c as varying exclusively in $I := \{m(x)\}_{x\in E}$.

In any case, assumption (H3) allows to apply Proposition 2.4 as well to $\boldsymbol{N}$ and E_c^0, as to $\hat{\boldsymbol{N}}$ and E_c^∞, in order to conclude that $\hat{N}(E_c^0) = 0$ or $N(E_c^0) < \infty$, and $N(E_c^\infty) = 0$ or $\hat{N}(E_c^\infty) < \infty$, almost surely for any $c \in I$. This means that all the trajectories in $\Omega^\star(E)$ are such that $N(x) = 0$ when $m(x) < c_\star$ and $N(x) = 1$ when $m(x) > c^\star$, where

$$c_\star = \min\{m(x) \mid N(x)=1\} \quad \text{and} \quad c^\star = \max\{m(x) \mid N(x)=0\}$$

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are two finite random variables. Between $c_\star$ and $c^\star$ there is a (perhaps vanishing) instability period, loosely speaking consisting of the values $N(x) \in \{0,1\}$ included between the first $N(x)=1$ and the last $N(x)=0$ (although this one can precede the former; i.e., $c^\star \leqslant c_\star$ may happen, as well as $c_\star < c^\star$). The only additional trajectories, not responding to this pattern, are those in $\Omega_0(E) = \{N(E)=0\}$ and $\Omega_1(E) = \{\hat{N}(E)=0\}$.

This description of the trajectories of the random selection process allows to conclude:

Proposition 3.3. *Given $c \in I$, let $\Delta_c := N(E_c^0) - \hat{N}(E_c^\infty)$. If $c' < c$ are two consecutive values in I, then*

$$\Delta_c - \Delta_{c'} = \#\{x \in E \mid m(x) = c\}$$

for any trajectory in $\Omega^\star(E)$. Thus, in $\Omega^\star(E)$, Δ_c grows from $-\infty$ to $+\infty$ with deterministic increments along a trajectory determined by a single random choice of some Δ_c. Meanwhile, $\Delta_c \equiv -\infty$ in $\Omega_0(E)$, and $\Delta_c \equiv +\infty$ in $\Omega_1(E)$.

Since in $\Omega^\star(E)$ we must have $N(E_c^0) < \infty$ and $\hat{N}(E_c^\infty) < \infty$ for each $c > 0$, we can consider the joint distribution of $(N(E_c^0), \hat{N}(E_c^\infty))$:

$$\gamma_c(u,v) := \mathrm{P}\{N(E_c^0) = u,\ \hat{N}(E_c^\infty) = v\} \quad (u, v \geqslant 0)$$

that may need to be completed with $\gamma_c(+\infty, 0) := \mathrm{P}(\Omega_1(E))$ and $\gamma_c(0,+\infty) := \mathrm{P}(\Omega_0(E))$.

Now, for any $c_1 < c_2 \in I$, let $C := \{x \in E \mid c_1 < m(x) \leqslant c_2\}$, and define $n := \#C$. Assuming that $u_2 - u_1 + v_1 - v_2 = n$, we can compute

$$\mathrm{P}\{N(E_{c_1}^0) = u_1,\ \hat{N}(E_{c_1}^\infty) = v_1,\ N(E_{c_2}^0) = u_2,\ \hat{N}(E_{c_2}^\infty) = v_2\}$$

in two ways:

$$\gamma_{c_1}(u_1,v_1)\mathrm{P}\{N(C) = u_2 - u_1 \mid \hat{N}(E_{c_1}^\infty) = v_1\} = \gamma_{c_1}(u_1,v_1)\frac{\hat{M}_{n-u_2+u_1}(C)\hat{M}_{v_2}(E_{c_2}^\infty)}{\hat{M}_{v_1}(E_{c_1}^\infty)},$$

$$\gamma_{c_2}(u_2,v_2)\mathrm{P}\{N(C) = u_2 - u_1 \mid \hat{N}(E_{c_2}^0) = u_2\} = \gamma_{c_2}(u_2,v_2)\frac{M_{u_2-u_1}(C)M_{u_1}(E_{c_1}^0)}{M_{u_2}(E_{c_2}^0)}.$$

Using (2.2) and putting $w := u_1 - v_1$, the equality of the second members yields

$$\frac{\gamma_{c_2}(v_2+w+n, v_2)}{M_{v_2+w+n}(E_{c_2}^0)\hat{M}_{v_2}(E_{c_2}^\infty)} = \frac{\gamma_{c_1}(v_1+w, v_1)}{M_{v_1+w}(E_{c_1}^0)\hat{M}_{v_1}(E_{c_1}^\infty)} \cdot \frac{1}{\prod_{x\in C} m(x)} \tag{3.7}$$

and, since the first member does not depend on v_1 nor the second member on v_2, we may rewrite (3.7) as

$$\varphi_{c_2}(w+n) = \varphi_{c_1}(w)\frac{1}{\prod_{x\in C} m(x)}, \tag{3.8}$$

with $\varphi_c(w)$ being a function of $w \in \mathbb{Z}$, that allows to write

$$\gamma_c(v+w, v) = \varphi_c(w) M_{v+w}(E_c^0)\hat{M}_v(E_c^\infty).$$

More precisely, the distribution $\pi_c(w) := \mathrm{P}\{N(E_c^0) - \hat{N}(E_c^\infty) = w\}$ will be given by

$$\pi_c(w) = \sum_{v=0}^{\infty} \gamma_c(v+w, v) = \varphi_c(w)\sum_{v=0}^{\infty} M_{v+w}(E_c^0)\hat{M}_v(E_c^\infty) \tag{3.9}$$

(where the terms with $v + w < 0$ vanish) and

$$\gamma_c(v+w, v) = \pi_c(w)\frac{M_{v+w}(E_c^0)\hat{M}_v(E_c^\infty)}{\sum_{v=0}^{\infty} M_{v+w}(E_c^0)\hat{M}_v(E_c^\infty)}. \tag{3.10}$$

But, according to Proposition 3.3, if $c_1 < c_2$ and $\nu_{c_1,c_2} := \#\{x \in E \mid c_1 < m(x) \leqslant c_2\}$, it will be

$$\pi_{c_2}(w + \nu_{c_1,c_2}) = \pi_{c_1}(w).$$

This conclusion may be confirmed directly from (3.8) and (3.9). Thus, if $c_0 \in I$ is any fixed value and $\pi(w) := \pi_{c_0}(w)$, all the distributions π_c may be obtained from π as

$$\pi_c(w) = \begin{cases} \pi(w - \nu_{c_0,c}) & \text{if } c > c_0, \\ \pi(w + \nu_{c,c_0}) & \text{if } c < c_0. \end{cases} \tag{3.11}$$

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Furthermore, any distribution π on $\mathbb{Z}$ gives a system of bidimensional distributions $\gamma_c(u,v)$ when (3.11) is replaced in (3.10) and, of course, (3.7) is verified. In fact, π may be any probability distribution on $\mathbb{Z}\cup\{\pm\infty\}$ such that $\pi(-\infty) := \mathrm{P}(\Omega_0(E))$ and $\pi(+\infty) := \mathrm{P}(\Omega_1(E))$.

In summary, a single distribution π on $\mathbb{Z}\cup\{\pm\infty\}$ determines all the marginal distributions of $\{N(B)\}_{B\in\mathcal{B}}$, and there exists an isomorphism between the convex set Π of distributions on $\mathbb{Z}\cup\{\pm\infty\}$ and the convex set $\mathcal{P}$ of marginal distributions of a random selection process. In this way, the extremes of $\mathcal{P}$ correspond to the causal distributions of Π. For these, Δ_{c_0} has a fixed value $w\in\mathbb{Z}$, as well as any

$$\Delta_c = \begin{cases} w+\nu_{c_0,c} & \text{if } c>c_0, \\ w-\nu_{c,c_0} & \text{if } c<c_0 \end{cases} \quad \text{and} \quad \gamma_c(v+\Delta_c,v) = \frac{M_{v+\Delta_c}(E_c^0)\hat{M}_v(E_c^\infty)}{\sum_{v=0}^\infty M_{v+\Delta_c}(E_c^0)\hat{M}_v(E_c^\infty)}.$$

In order to compute the associated marginal distribution of $N(B)$, for $B\in\mathcal{B}$, we can always choose a value of c larger than $\max_{x\in B} m(x)$, and we will thus have

$$\mathrm{P}\big\{N(B)=s\big\} = M_s(B)\frac{\sum_{v=0}^\infty M_{v+\Delta_c-s}(E_c^0-B)\hat{M}_v(E_c^\infty)}{\sum_{v=0}^\infty M_{v+\Delta_c}(E_c^0)\hat{M}_v(E_c^\infty)} =: \mathcal{H}^\star(w;s,B) \tag{3.12}$$

where $\mathcal{H}^\star(w;s,B)$ depends on $w\in\mathbb{Z}$ but it must be, however, independent of c (although the expression will be different if $c<\min_{x\in B} m(x)$ and even more complicated when $c\in\{m(x)\}_{x\in B}$). The obtained result is formulated in the following theorem.

Theorem 3.4. *Let* $\boldsymbol{N}$ *be a random selection process on* (E,m) *and assume that* (H3) *is verified. Then, the marginal distributions of* $\{N(B)\}_{B\in\mathcal{B}}$ *are given by*

$$\mathrm{P}\big\{N(B)=s\big\} = \sum_{w\in\mathbb{Z}} \pi(w)\mathcal{H}^\star(w;s,B) + \pi(-\infty)\mathbf{I}_{\{s=0\}} + \pi(+\infty)\mathbf{I}_{\{s=\#B\}} \tag{3.13}$$

where $\pi(w)$ *is any probability distribution on* $\mathbb{Z}\cup\{\pm\infty\}$. *These marginal distributions are, thus, mixtures of the extreme distributions* $\{\mathcal{H}^\star(w;s,B)\}_{w\in\mathbb{Z}\cup\{\pm\infty\}}$ *given in* (3.12).

Once again, the "generalized binomial" distributions may be recovered by an appropriate choice of the $\pi(w)$. Concretely, for any $\theta\in(0,\infty)$, let

$$\varphi_c(\theta,w) := \frac{\theta^w}{\prod_{x\in E_c^0}[1+\theta m(x)]\prod_{x\in E_c^\infty}[1+\hat{m}(x)/\theta]}.$$

It is easy to see that (3.8) is satisfied and the corresponding distributions $\gamma_c(u,v)$ are given by

$$\frac{\theta^{u-v}M_u(E_c^0)\hat{M}_v(E_c^\infty)}{\prod_{x\in E_c^0}[1+\theta m(x)]\prod_{x\in E_c^\infty}[1+\hat{m}(x)/\theta]};$$

then, assuming that $c>\max_{x\in B} m(x)$, a simple calculus shows that

$$\mathrm{P}\big\{N(B)=s\big\} = \sum_{u=s}^\infty\sum_{v=0}^\infty \gamma_c(u,v)\frac{M_s(B)M_{u-s}(E_c^0-B)}{M_u(E_c^0)} = \frac{M_s(B)\theta^s}{\prod_{x\in B}[1+\theta m(x)]}.$$

The same result may be obtained starting from any value of c. In order to reproduce exactly (1.3) it only remains to replace θ with $\xi/1-\xi$ and to choose $\xi\in[0,1]$ with distribution F.

Besides, let A be any denumerable set in E. If $\{m(x)\}_{x\in A}$ has only one limit point, 0 or ∞, then (3.5) or (3.6) holds, respectively. In case that A has both limit points, fix an arbitrary $c\in I$ in order to consider $A^0 := A\cap E_c^0$ and $A^\infty := A\cap E_c^\infty$, $A_n^0 := A_n\cap E_c^0$, $A_n^\infty := A_n\cap E_c^\infty$, and $\kappa(n) := \#A_n^\infty$. Observe that $N(A_n^0)-\hat{N}(A_n^\infty) = N(A_n)-\kappa(n)$. Moreover, for large enough n, $N(A_n^0)$ equals $N(A^0)$, and $\hat{N}(A_n^\infty)$ equals $\hat{N}(A^\infty)$. Therefore

$$\mathcal{G}_\infty^A = \sigma\big(N\big(A^0\big)-\hat{N}\big(A^\infty\big)\big).$$

Note that when c is replaced with $c'<c$, the variation of $\Delta_c^A := N(A^0)-\hat{N}(A^\infty)$ equals the constant value $\nu_{c',c}^A := \#\{x\in A \mid c'<m(x)\leqslant c\}$; consequently $\mathcal{G}_\infty^A$ remains unchanged.

Furthermore, given any finite $B\subset A$, for the calculus of $Z_\infty^A(s,B)$ we may always assume that c has been fixed greater than $\max_{x\in B} m(x)$; then

$$Z_\infty^A(s,B) = M_s(B)\left[\frac{\sum_{v=0}^\infty M_{v+\delta-s}(A^0-B)\hat{M}_v(A^\infty)}{\sum_{v=0}^\infty M_{v+\delta}(A^0)\hat{M}_v(A^\infty)}\right]_{\delta=N(A^0)-\hat{N}(A^\infty)}. \tag{3.14}$$

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This conclusion is not too different from (3.5) and (3.6) because, when $\{m(x)\}_{x\in A}$ is bounded above, the value of c may be fixed greater than the bound in order to have $A^\infty = \emptyset$. Then, both sums in (3.14) reduce to the term $v = 0$, and (3.5) is recovered. A symmetric procedure gives (3.6) if the weights in A are bounded below.

This argument shows, more generally, that the results obtained under (H1) and (H2) in Section 3.1 are, in some way, particular cases of Theorem 3.4. Indeed, if $I = \{m(x)\}_{x\in E}$ is bounded above, the judicious choice of c is to take it greater than the bound, so that $E_c^\infty = \emptyset$ and $\mathcal{H}^\star(w; s, B)$ reduces to $\mathcal{H}(w; s, B)$. In the same way, $\mathcal{H}^\star(w; s, B)$ reduces to $\hat{\mathcal{H}}(w; s, B)$ when I is bounded below and c is taken lower than its bound. Actually, the peculiarity of the cases (H1) and (H2) is that, for each $c > 0$, Δ_c has a range in $\mathbb{Z}$ bounded below or above, respectively; a convenient choice of c reduces this range to $\mathbb{N}$. On the contrary, when (H3) is in force, the range of Δ_c is $\mathbb{Z}$ for any choice of $c > 0$, and no further simplification of (3.12) is possible.

3.3. The rejection function

Finally, a remark concerning the rejection function $p_0(x) := \mathrm{P}\{N(x) = 0\}$ is in order. It has been shown in [10, Lemmas 3.2 and 3.4] that $p_0(x)$ is a function of $m(x)$, and that it determines the distribution of the random selection process provided that $m(x)$ is an injective function on E. Furthermore, it may be characterized as a totally decreasing function on $I := \{m(x)\}_{x\in E}$, as defined in [11, Section 4].

When I contains a sequence of numbers m_j such that $\sum_{j=1}^\infty m_j = \infty$ or $\sum_{j=1}^\infty 1/m_j = \infty$, it was proved in [11, Theorem 4.2] that, for some probability distribution F on $[0, 1]$,

$$p_0(m) = \int_0^1 \frac{1-\xi}{1-\xi+\xi m} F(d\xi). \tag{3.15}$$

Nevertheless, in the present context, as $m(E_c^0) < \infty$ and $\hat{m}(E_c^\infty) < \infty$ for any $c \in I$, $p_0(m)$ may be any convex mixture of the extremes

$$\frac{\sum_{v=0}^\infty M_{v+w}(I_c^0 - m)\hat{M}(I_c^\infty)}{\sum_{v=0}^\infty M_{v+w}(I_c^0)\hat{M}_v(I_c^\infty)} \quad (w \in \mathbb{Z})$$

that, when I is bounded above or below, can be simplified to

$$\frac{M_u(I-m)}{M_u(I)} \quad (u \in \mathbb{N}) \quad \text{or} \quad \frac{\hat{M}_v(I-m)}{\hat{M}_v(I)} \quad (v \in \mathbb{N}),$$

respectively. In any case, all the extremes depend on all the values in I, though (3.15) may be recovered by means of an adequate choice of the mixture coefficients.

4. An example

We conclude by proposing an example of a random selection process $\boldsymbol{N}$ with non-constant weight function m (that is, the indicator random variables $N(x)$ are not exchangeable; recall Proposition 1.1).

Suppose that a basin is filled at a randomly chosen and unobservable constant flow of z particles per time unit. We assume that filling takes place at times labeled $t = 1, 2, \ldots$. Our goal is to estimate the flow z. To this end, we put into the basin some "special" and distinguishable particles. More precisely, we suppose that, at time $t \geqslant 1$, we put $\Delta_t \geqslant 0$ particles into the basin, so that, at time $t \geqslant 1$, there are zt particles coming from the filling, and

$$y(t) = \Delta_1 + \cdots + \Delta_t$$

special particles.

At each time $t \geqslant 1$ we draw (with replacement) a particle from the basin, and, upon this sequence of observations, we must estimate the flow z. For each $t = 1, 2, \ldots$, let $N(t) := 1$ if the particle drawn at time t is among the $y(t)$ special ones that we threw into the basin, and let $N(t) := 0$ otherwise. (Let us mention that, if z is assumed to have a continuous distribution, then we can think of the special particles as volumes added to the basin, and the results below remain valid.)

We can verify that the conditions (C1) and (C2) in the definition of a random selection process hold. As an illustration, let us check the condition (C2). Let B be a finite set of time labels and fix $t \in B$. Then, letting F be the distribution of z,

$$\mathrm{P}\big\{N(t) = 1 \mid N(B) = 1\big\} = \frac{\int \frac{y(t)}{zt} \prod_{x\in B} \frac{zx}{y(x)+zx} F(dz)}{\sum_{w\in B} \int \frac{y(w)}{zw} \prod_{x\in B} \frac{zx}{y(x)+zx} F(dz)} = \frac{y(t)/t}{\sum_{w\in B} y(w)/w},$$

so that (C2) holds for the weight function $m(t) := y(t)/t$ for $t = 1, 2, \ldots$. The condition (C1) is verified similarly.

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Consequently, the random variables $\{N(t)\}_{t\geqslant 1}$ are a random selection process with weight function

$$m(t) = \frac{y(t)}{t} = \frac{\Delta_1 + \cdots + \Delta_t}{t},$$

for $t \geqslant 1$. Hence, they are not exchangeable, except for the case of a constant Δ_t.

Regarding the estimation of the flow z, there are some cases of special interest. Suppose that $y(t)$, for $t \geqslant 1$, is of the form at^{α}, where $a > 0$ and $\alpha \geqslant 0$ are given constants, so that the weight function is $m(t) = at^{\alpha-1}$, for $t \geqslant 1$. Then we distinguish the following alternatives (in which the case $\alpha = 1$, corresponding to exchangeability, is discarded):

Case $0 \leqslant \alpha < 1$. The weight function is $m(t) = at^{\alpha-1}$, for $t \geqslant 1$, so that a De Finetti theorem holds (recall the condition (H1) in Section 3). In addition, by the strong law of large numbers in Theorem 2.1 in [11] (see (1.5)),

$$\frac{a\sum_{t=1}^{n} t^{\alpha-1}}{\sum_{t=1}^{n} N(t)}$$

is an almost surely consistent estimator of z as $n \to \infty$. In particular, if $\alpha = 0$ then we put a particles into the basin at time $t = 1$, while $\Delta_2 = \Delta_3 = \cdots = 0$. In this case,

$$\lim_{n\to\infty} \frac{a \log n}{\sum_{t=1}^{n} N(t)} = z$$

with probability one.

Case $1 < \alpha \leqslant 2$. Now, the weight function verifies the condition (H2). Hence, the De Finetti theorem still holds and, by [11, Theorem 2.3],

$$\lim_{n\to\infty} \frac{a(n - \sum_{t=1}^{n} N(t))}{\sum_{t=1}^{n} t^{1-\alpha}} = z$$

with probability one. If $\alpha = 2$, then

$$\lim_{n\to\infty} \frac{a(n - \sum_{t=1}^{n} N(t))}{\log n} = z$$

almost surely.

Case $\alpha > 2$. The weights $m(t)$ are such that $m(t) \uparrow \infty$ as $t \to \infty$, though

$$\sum_{t=1}^{\infty} 1/m(t) < \infty.$$

In this case, even if the random variables $N(t)$ are conditionally independent given z, we do not obtain a law of large numbers because $\sum_t (1 - N(t))$ is finite with probability one (that is, we draw a finite number of particles coming from the filling at flow z). The interpretation is that, for $\alpha > 2$, we have put so many special particles into the basin that the ones whose flow is to be estimated become asymptotically negligible.

5. Conclusions

We may summarize the results proved here and in [11] in a final result that completely describes all the random selection processes on (E, m), for any space (E, m) (though the trivial case of a finite E will be omitted).

Theorem 5.1. *Let* $\boldsymbol{N}$ *be a random selection process on* (E, m). *If* (E, m) *verifies some of the hypotheses* (A1), (A2) or (A3), *then*:

(DF) *There exists a random variable* $\xi \in [0, 1]$ *such that* $\{N(x)\}_{x\in E}$ *are conditionally independent given* ξ, *with distributions* $\mathrm{P}\{N(x) = 1 \mid \xi\} = \xi m(x)/(1 - \xi + \xi m(x))$ *for all* $x \in E$.

Otherwise, (H1), (H2) *or* (H3) *are verified, and then* (DF) *may not hold. In fact, the marginal distributions of* $\boldsymbol{N}$, *as described in Theorems* 3.1, 3.2 *and* 3.4, *are mixtures of the discrete families of distributions*

$$\left\{\mathcal{H}(u; s, B)\right\}_{u\in\mathbb{N}\cup\{\infty\}}, \qquad \left\{\hat{\mathcal{H}}(v; s, B)\right\}_{u\in\mathbb{N}\cup\{\infty\}}, \quad \text{or} \quad \left\{\mathcal{H}^{\star}(w; s, B)\right\}_{w\in\mathbb{Z}\cup\{\pm\infty\}},$$

respectively. Thus, (DF) *only holds for particular choices of* $\gamma(u)$, $\hat{\gamma}(v)$ *and* $\pi(w)$, *respectively.*

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In the final section of [10] we called *fine* a sequence of weights $\{m_i\}$ such that a random selection process on $(\mathbb{N}, \{m_i\})$ has necessarily marginal distributions of the form (1.3). We wondered *if nonfine infinite sequences could exist* and then *what are the corresponding families of marginal distributions*. We now know that any sequence verifying (H1), (H2), or (H3) is nonfine and Theorems 3.1, 3.2, 3.4 describe the set of all families of marginal distributions.

Certainly, the setting of random selection processes may be extended to more general situations, allowing to assign to each point in E a random index i from a finite set $L = \{0, 1, \ldots, \ell\}$ with similar rules to those proposed in (C1) and (C2) in Section 1. The main difference is that several weights $m_i(x)$ may regulate the allocation of the index i within a finite set $B \subset E$, once it is known that B contains no further nonzero indexes. The precise formulation of this kind of *random assignment processes* and a De Finetti-type result, under an assumption close to (A1), is described in [9]. Of course, a complete analysis—in the line of that achieved in Theorem 5.1—must be more involved for random assignment processes. Beyond, the possibility of considering a denumerable index set L, or even a measurable index set $(L, \mathcal{L})$, as done in [5] for the exchangeable case, remains open.

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